

Lecture 5: Inference for Subpopulation Means and Differences

September 6, 2023

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- ▶ But we also demonstrated that we could accurately predict how close we might be by using confidence intervals.
- ▶ We also saw that our sample difference in subpopulation proportions was likely to be somewhat close to the population difference in subpopulation proportions.
- ▶ Again, we also demonstrated that we could predict how close we might be by using confidence intervals.
- ▶ We used a bootstrapping approach to confidence intervals and showed how bootstrapping differences allowed us to generate numbers that were close to the numbers in a regression table.

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- ▶ Returning to the CPS example, we will consider college degree incomes and the difference between college degree and non-college degree incomes.
- ▶ Even when the overall sample size is fixed (2258 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- ▶ This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

Income and 4-year degree from the CPS

some review of the CPS regression from lecture 1 including the subpopulation means

CPS Sampling Distribution

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- ▶ We will also never know the subpopulation income standard deviations, which we will denote σ_1 for those with a 4-yr degree and σ_0 for those without.
- ▶ We also will not be able to generate a sampling distribution, but we still want to know things about the sampling distribution.

Sampling Distribution for 4-yr degree Mean Income

Let the variable X take the value 1 if Black and 0 if non-Black

i	1 Y_1, X_1	2 Y_2, X_2	...	625 Y_{2258}, X_{2258}	$\bar{Y}_{2258}$	$\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i}$
	37,006,0	108,811,1	...	49,001,0	135,406	150,000?
$f(\cdot)$			...		$N(\mu, \frac{\sigma^2}{n})$	
$E(\cdot)$			...		μ	
$V(\cdot)$			...		$\frac{\sigma^2}{n}$	

- ▶ Although we can't observe the population or the sampling distribution, we can mathematically derive certain facts about the sampling distribution.
- ▶ We will focus on the three red question marks.

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1	37,006,0	108,811,1	...	49,001,0	135,406	150,000?
2	??	??	...	??	??	??
$f(\cdot)$			...		$N(\mu, \frac{\sigma^2}{n})$	
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1	37,006,0	108,811,1	...	49,001,0	135,406	150,000?
2	?,?	?,?	...	?,?	?,?	?,?
3	?,?	?,?	...	?,?	?,?	?,?
4	?,?	?,?	...	?,?	?,?	?,?
5	?,?	?,?	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?,?	?,?	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$			...		$N(\mu, \frac{\sigma^2}{n})$	
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$f(\cdot)$			...		$N(\mu, \frac{\sigma^2}{n})$	?
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Property #1: *The sampling distribution of the subsample mean is approximately normal when $\sum_{i=1}^{2258} X_i$ is large.*

- ▶ The approximate normality result comes from a central limit theorem (CLT).
- ▶ We know that $\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i} \sim_{approx} N(?, ?)$, but still don't know the mean or the variance of this distribution.

Property #2: *The subsample mean is unbiased for the subpopulation mean.*

Part 1 of Proof (condition on $X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}$):

$$\begin{aligned} & E\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}\right] \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} E\left[\sum_{i=1}^{2258} Y_i X_i \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}\right] \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} \sum_{i=1}^{2258} E[Y_i X_i \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}] \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} \sum_{i=1}^{2258} x_i E[Y_i \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}] \end{aligned}$$

Property #2: *The subsample mean is unbiased for the subpopulation mean.*

Part 2 of Proof (condition on $X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}$):

$$\begin{aligned} & \frac{1}{\sum_{i=1}^{2258} x_i} \sum_{i=1}^{2258} x_i E[Y_i | X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}] \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} \sum_{i=1}^{2258} x_i E[Y_i | X_i = x_i] \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} \sum_{i=1}^{2258} x_i [x_i \mu_1 + (1 - x_i) \mu_0] \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} \sum_{i=1}^{2258} x_i \mu_1 \\ &= \frac{1}{\sum_{i=1}^{2258} x_i} \mu_1 \sum_{i=1}^{2258} x_i = \mu_1. \end{aligned}$$

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Part 3 of Proof:

$$\begin{aligned} & E\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i}\right] \\ &= E_{\mathbf{X}}\left[E\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}\right]\right] \\ &= E_{\mathbf{X}}[\mu_1] \\ &= \mu_1 \end{aligned}$$

Property #3: The *sampling variance* of the subsample mean,

Part 1 of Proof:

$$\begin{aligned} & V\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i}\right] \\ &= E_{\mathbf{X}}\left[V\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}\right]\right] \\ &+ V_{\mathbf{X}}\left[E\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}\right]\right] \\ &= E_{\mathbf{X}}\left[V\left[\frac{\sum_{i=1}^{2258} Y_i X_i}{\sum_{i=1}^{2258} X_i} \mid X_1 = x_1, X_2 = x_2, \dots, X_{2258} = x_{2258}\right]\right] \\ &+ V_{\mathbf{X}}[\mu_1] \end{aligned}$$

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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
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$f(\cdot)$		...	...		$N(\mu, \frac{\sigma^2}{n})$	
$E(\cdot)$		...	...		μ	
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?,?	...	...	?,?	?,?	?,?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$		...	...		$N(\mu, \frac{\sigma^2}{n})$	$N(\mu_1, \sigma_1^2 E(\frac{1}{\sum_{i=1}^{2258} x_i}))$
$E(\cdot)$		...	...		μ	μ_1
$V(\cdot)$		...	...		$\frac{\sigma^2}{n}$	$\sigma_1^2 E(\frac{1}{\sum_{i=1}^{2258} x_i}))$

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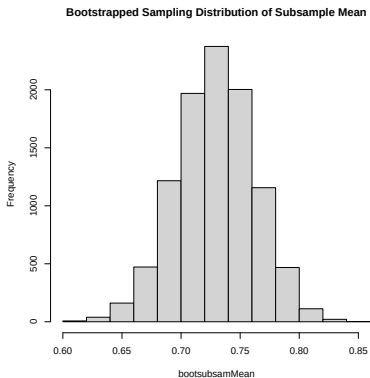
Simulating the Bootstrapped Sampling Distribution

i	1	2	...	625	$\bar{y}_{2258}$	$\bar{y}_1 = \frac{\sum_{i=1}^{2258} y_i x_i}{\sum_{i=1}^{2258} x_i}$
y_i	36,006	108,811	...	49,001	135,406	150,000
x_i	0	1	...	0		

Starting with the sample (first row of sampling distribution), each row of the bootstrapped sampling distribution is generated as the following:

```
bootind = sample(1:2258, size=2258, replace=TRUE)
dfboot = dfsam[bootind,]
subsamMean = mean(dfboot$y[dfboot$x == 1])
```

```
B = 10000  
bootsubsamMean = rep(NA,B)  
for( b in 1:B ){  
  bootind = sample(1:625,size=625,replace=T)  
  dfboot = dfsam[bootind,]  
  bootsubsamMean[b] = mean(dfboot$y[dfboot$x==1])  
}
```



Plug-in CI

Estimates

- ▶ $\mu_1: \bar{y}_1 = \frac{\sum_{i=1}^{2258} y_i x_i}{\sum_{i=1}^{2258} x_i}$
- ▶ $\sigma_1^2: \frac{\sum_{i=1}^{2258} (y_i - \bar{y}_1)^2 x_i}{\sum_{i=1}^{2258} x_i - 1}$
- ▶ $E\left(\frac{1}{\sum_{i=1}^{2258} x_i}\right): \frac{1}{\sum_{i=1}^{2258} x_i}$

95% CI:

$$\bar{y}_1 \pm 1.96 \cdot \frac{1}{\sum_{i=1}^{2258} x_i} \cdot \frac{\sum_{i=1}^{2258} (y_i - \bar{y}_1)^2 x_i}{\sum_{i=1}^{2258} x_i - 1}$$

Sampling Distribution for Difference in Mean Income

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	37,006,0	$\dots$	$\dots$	49,001,0	150,000?	25,000?
$f.(\cdot)$		$\dots$	$\dots$		$N(\mu, \frac{\sigma^2}{n})$ μ $\frac{\sigma^2}{n}$	
$E.(\cdot)$		$\dots$	$\dots$			
$V.(\cdot)$		$\dots$	$\dots$			

- We still don't know μ_1, σ_1^2 , or $E(\frac{1}{\sum_{i=1}^{2258} x_i})$, so we have to bootstrap or estimate them.

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1	37,006,0	...	...	49,001,0	150,000?	25,000?
2	?,?	...	...	?,?	?,?	?,?
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5	?,?	...	...	?,?	?,?	?,?
:	:	:	:	:	:	:
1 mil	?,?	...	...	?,?	?,?	?,?
:	:	:	:	:	:	:
:	:	:	:	:	:	:
$f.(.)$		...	...		$N(\mu, \frac{\sigma^2}{n})$	
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1	37,006,0	...	...	49,001,0	150,000?	25,000?
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4	?,?	...	...	?,?	?,?	?,?
5	?,?	...	...	?,?	?,?	?,?
...	...	...	...	...	...	...
1 mil	?,?	...	...	?,?	?,?	?,?
...	...	...	...	...	...	...
...	...	...	...	...	...	...
$f(\cdot)$		...	...		$N(\mu, \frac{\sigma^2}{n})$	$N(\cdot, \cdot)$
$E(\cdot)$		...	...		μ	$\mu_1 - \mu_0$
$V(\cdot)$		...	...		$\frac{\sigma^2}{n}$	$\sigma_1^2 E(\frac{1}{\sum_{i=1}^{2258} X_i}) + \sigma_0^2 E(\frac{1}{\sum_{i=1}^{2258} (1-X_i)})$

- We still don't know μ_1 , σ_1^2 , or $E(\frac{1}{\sum_{i=1}^{2258} X_i})$, so we have to bootstrap or estimate them.